Tyler Stoeger

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EDUCATION

University of Illinois Chicago — **Chicago, IL**
Ph.D. in Economics — Expected May 2027
M.A. in Applied Economics — May 2024
CMEP Exchange Scholar, University of Chicago Harris School of Public Policy — *2024*
Courses taken: Energy and Environmental Economics I and III

Purdue University Northwest — **Hammond, IN**
Bachelor of Science in Business, Major in Finance — May 2022
Minors in Economics and Applied Mathematics
GPA: 4.00/4.00; Highest Distinction

RESEARCH FIELDS

Environmental, Resource, and Energy Economics; Public Economics; Applied Microeconomics

JOB MARKET PAPER

Behavioral Responses to Air Quality Information: Implications for Optimal Policy
2026 Graduate Silvia Saffrin Award for Excellence in the Third-Year Paper

Abstract. Air quality alerts can improve welfare by helping individuals avoid pollution exposure. However, lowering the alert threshold creates a tradeoff: issuing more warnings can encourage avoidance on additional days, but may also change the effectiveness of subsequent alerts. I develop a model of optimal air quality alert policy that accounts for these history-dependent behavioral responses. Determining the optimal threshold requires understanding both how alerts affect outdoor activity and how responsiveness varies with recent alert exposure. I estimate these effects using a fuzzy regression discontinuity design that exploits the jump in alert issuance at a forecasted air quality index (AQI) of 101, combining a panel of air quality forecasts, alert issuances, and high-frequency foot traffic data from six U.S. metropolitan areas. Near the threshold, receiving an alert reduces daily visits to outdoor recreational points of interest by approximately 9 percent. However, the effect depends strongly on recent alert history. There is no detectable response when no alerts have occurred in the previous seven days; alerts reduce visits by roughly 16 percent following one or two recent alerts, but the effect attenuates after three or more alerts. I then use these estimates to calibrate the model and conduct counterfactual simulations using data from New York City. The simulations place the welfare-maximizing alert threshold at an AQI of approximately 80, below the current threshold of 101. These results show that optimal warning policy depends not only on the risks communicated by alerts, but also on how repeated warnings shape the effectiveness of subsequent alerts.

WORKING PAPERS

Air Quality Forecasts in Practice — **Draft expected Nov. 2026**
This paper examines systematic differences between forecasted and realized air quality across U.S. metropolitan areas. Preliminary results indicate that forecasts overpredict realized AQI by approximately 9 points on average, with substantial dispersion in realized AQI at higher forecast levels. Forecasts and realized conditions also exhibit different patterns of serial correlation. The analysis investigates potential explanations for these discrepancies, including meteorological conditions and features of the forecasting process.

REFERENCES

Matthew Tarduno (Dissertation Chair) — **tarduno@uic.edu**
Assistant Professor of Economics, University of Illinois Chicago

Benjamin Feigenberg — **bfeigenb@uic.edu**
Professor of Economics, University of Illinois Chicago

Ben Ost — **bost@uic.edu**
Professor of Economics, University of Illinois Chicago

HONORS & AWARDS

Graduate Silvia Saffrin Award, University of Illinois Chicago 2026
Chancellor's Medallion, Purdue University Northwest 2022
Awarded to graduates earning the highest GPA in their academic college

RESEARCH GRANTS

Award for Graduate Research, UIC Graduate College 2025
$2,900 in research support
Undergraduate Research Grant, Purdue University Northwest Fall 2021
Undergraduate Research Grant, Purdue University Northwest Spring 2021

PRESENTATIONS

2026
Southern Economic Association Annual Meeting, Graduate Student Session, Houston, TX Forthcoming
2025
University of Chicago, Environmental and Energy Economics Graduate Student Workshop December

RESEARCH EXPERIENCE

University of Illinois Chicago **Fall 2024; Summer 2025**
Research Assistant to Matthew Tarduno
Conducted literature reviews, collected and cleaned data, performed preliminary empirical analysis, and edited manuscripts for two research projects.
Purdue University Northwest **Jan. 2021 – May 2022**
Research Assistant to Amlan Mitra
Conducted literature reviews, collected and cleaned data, and supported preliminary empirical analysis across faculty and funded undergraduate research projects.

TEACHING EXPERIENCE

University of Illinois Chicago
Instructor of Record
ECON 300 - Econometrics Summer 2024; Spring 2025; Fall 2025
Designed and delivered lectures, assignments, examinations, and applied statistical-programming labs; supervised a teaching assistant.
Teaching Assistant
ECON 300 - Econometrics Summer 2023; Spring 2024; Fall 2026
ECON 534 - Econometrics I Fall 2023; Fall 2024
ECON 121 - Principles of Macroeconomics Fall 2022; Spring 2026
ECON 120 - Principles of Microeconomics Spring 2023; Fall 2023
ECON 220 - Microeconomics: Theory and Applications Spring 2023
ECON 510 - Microeconomic Theory II Spring 2024
Graduate Assistant
ECON 516 - Development Economics Spring 2026

ADDITIONAL TRAINING

Berkeley/Sloan Summer School in Environmental and Energy Economics, UC Berkeley 2025

PROFESSIONAL MEMBERSHIPS

Association of Environmental and Resource Economists (AERE); American Economic Association (AEA); Southern Economic Association (SEA); Illinois Economic Association (IEA)

TECHNICAL SKILLS

Software: R, Stata, MATLAB, LaTeX, Git/GitHub, Excel.
Additional: Spatial data analysis, geographic data processing, and counterfactual simulation.